

## Multiple Choice (10 marks)

1. Radio telescopes are crucial for astronomical observations. Which one of these well-known radio telescopes has the largest parabolic antenna?
  - (a) Green Bank Telescope
  - (b) Arecibo Telescope
  - (c) Yevpatoria RT-70 Telescope
  - (d) Effelsberg Telescope
2. Which is the least likely cause of death?
  - (a) Being hit in the head by a bullet.
  - (b) Being hit by a small meteorite.
  - (c) Starvation during global winter caused by a major impact.
  - (d) Driving while intoxicated without wearing seatbelts.
3. Jupiter and the other jovian planets are sometimes called "gas giants." In what sense is this term misleading?
  - (a) The materials they are made of are not normally gaseous in everyday experience.
  - (b) They actually contain a significant fraction of non-gaseous matter.
  - (c) The materials that make up these planets are primarily in the form of a plasma not a gas.
  - (d) Actually it's a great description because these worlds are big and gaseous throughout.
4. Which of the following lists the ingredients of the solar nebula from highest to lowest percentage of mass in the nebula?
  - (a) hydrogen compounds ( $\text{H}_2\text{OCH}_4\text{NH}_3$ ) light gases (H He) metals rocks
  - (b) hydrogen compounds ( $\text{H}_2\text{OCH}_4\text{NH}_3$ ) light gases (H He) rocks metals
  - (c) light gases (H He) hydrogen compounds ( $\text{H}_2\text{OCH}_4\text{NH}_3$ ) metals rocks
  - (d) light gases (H He) hydrogen compounds ( $\text{H}_2\text{OCH}_4\text{NH}_3$ ) rocks metals
5. The name of the black hole in the center of our Milky Way is ...
  - (a) Altair A\*
  - (b) Alsephina A\*
  - (c) Fomalhaut A\*
  - (d) Sagittarius A\*

## True / False (10 marks)

*Answer True or False*

6. True or False: The Great Red Spot is a prominent feature found on the surface of the planet Saturn.
7. True or False: The altitude of the north celestial pole is equal to your latitude in the northern hemisphere.
8. True or False: The ecliptic is defined as the apparent path of the Sun across the celestial sphere over the course of a year.
9. True or False: Pluto has one medium-sized satellite, Charon, and two smaller satellites, Nix and Hydra.
10. True or False: Dark energy describes the radiation emitted by black holes and their interaction with surrounding matter.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the changes in the apparent altitude of Polaris as one travels north from the United States into Canada, and explain the underlying astronomical principles involved.
12. Discuss how the loss of Mars's magnetic field may have contributed to climatic changes on the planet, particularly in terms of atmospheric loss and its impact on surface conditions.

*End of Paper*